

2 선적 전 화물관리 Cargo Control before Loading

2.1 일반사항 (General)

선장은 선적 전 다음 각 항에 언급된 사항을 면밀히 조사하여 사전에 화물 손상을 입힐 우려가

있는 요소들을 미리 발견, 예방하여야 하며, 선창에 하자사항 발견 시 지체 없이 운항감독에 통보하여 결함부분을 수리 또는 시정 후 선적 작업을 착수해야 한다 (참고로 용선주는 또는 용선으로 화물을 운송하는 OPERATOR는 선박의 ON HIRE SURVEY시 BOARDING SURVEYOR로 하여금 화물안전 수송에는 별 영향을 끼치지 않으나 REDELIVERY시 문제점으로 대두될 수 있는 항목 “HOLD 내 FRAME 손상, DECK 상 HANDRAIL 손상 등”을 상세히 CHECK하여

RE-DELIVERY시 용선기간 중 일어난 DAMAGE가 아님을 증거로 남겨야 한다.)

The master should inspect the following items before loading cargo in order to prevent cargo damage by removing any possible causes of cargo damage and if finding defects in the cargo hold, report this to the Operation superintendent so that the cargo can be loaded after repairing these defects. (The master should remember that when on a hire survey, ship owners or the operators who transport cargo by chartered ships tend to let boarding surveyors check carefully the items such as the damage to the ship's frame, deck or handrails which cannot affect the safety of cargo transportation but can cause trouble when redelivering the ship in order to prove that the damage doesn't occur while chartering the ship).

2.2 선적 전 선체 점검 (Checking the Hull Prior to Loading)

다음 사항은 최초 선적항에서 선적 전 당해 항구에서 화주 또는 용선주 측의 BOARDING SURVEY가 검사하는 사항으로 화주 또는 용선주의 이익을 위하여 하자사항 발생 시 계약이행 거부, 시정 또는 수리 후 선적 착수하도록 하고 있으므로 선장은 선적 또는 용선이 거부되지 않도록 확인하여야 한다. 양식 “OPR-BCH-F03 화물 선적전 점검표”를 통해 점검 기록을 확인한다.

The following items will be inspected by a supercargo and a boarding surveyor of the ship's owner or charterer's choice in the initial port where cargo is loaded before loading the cargo and if there are defects in the ship, it is common to refuse to charter the ship until the defects are corrected. The master should check using by form “OPR-BCH-F03 Checklist before Cargo Loading” including following items in order to prevent the ship from being rejected.

2.2.1 선적 전 HOLD CLEANING (Hold cleaning prior to loading)

화물종류, 용선주의 요구사항 등을 확인하고, 전 항차 화물잔재 제거나 수세작업, BILGE BOX 소재 등의 작업 소요시간을 고려하여 수립하여야 한다.

OPR-BCH-F14 cargo hold cleaning check list 에 따라 청소를 실시한다.

To confirm the kind of next cargo and charterer's requirements etc.;

- The plan should be established with consideration to the time required for removing last cargo residues, washing, cleaning of bilge boxes, etc.
- Cargo hold cleaning will be performed according to OPR-BCH-F14 Cargo hold cleaning check list .

2.2.2 HOLD SWEEPING

전 항차 화물잔재 HOLD WASHING 시 BILGE PUMPING 으로 SUCTION 되지 않도록 HOLD WASHING 전 시행, 특히 시멘트와 같은 HEAVY CGO 일 경우 PUMP/VALVE SEALING 등의 손상 및 PIPE 폐쇄 초래할 개연성 상존하므로 가급적 이동용 PUMP 사용한다.
수압이 미치지 않는 높은 곳이나 ANGLE PROTECT 뒷면은 AIR BLOWING 을 통하여 제거 한다.

In case of carrying out hold washing to remove last cargo residues, the cargo residues should be removed before hold washing so that they are not sucked in during pumping out of bilges, particularly, in case of heavy cargoes, such as cement, there might be a possibility of damage to the pump and valve sealing, or clogging of the pipe. In such cases, a portable pump should be used if possible.

Cargo residues at high places, where hydraulic pressure cannot reach or behind angle brackets should be removed by air blowing

2.2.3 WATER WASHING

1) HOLD WASHING

사전 BILGE WELL 이 화물잔재로 막히지 않도록 조치한다.

BILGE BELL MOUTH ROSE BOX 장착 확인을 한다.

전 화물창이 완전히 CLEAN UP 되도록 충분한 WASHING 시행한다.

WASHING 중 BILGE LINE이 막히지 않도록 주기적으로 정지하고, CLEAN SEA WATER 재 WASHING을 한다.

종료시점 BILGE WELL 로 충분히 살수하여 BILGE LINE CLEAN UP 시행한다.

- To take measures in advance not to clog the bilge wells by cargo residues;
- To confirm whether the bilge bell mouth rose box is fitted or not;
- To carry out washing sufficiently to clean up all of the hold space;
- In order not to clog the bilge line during washing, the washing should be stopped regularly and the line should be washed with clean sea water; Upon completion of washing, the bilge line should be washed with clean sea water for sufficient time.

2) H/COVER, ACESSES, 화물창 고정식 소화장치 소제

Cleaning Hatch Cover, Access and Fixed Fire-fighting System for Cargo Hold

H/COVER FRAME, DRAIN CHANNEL, VENTILATOR, HOLD ACCESS 등 화물 잔재 제거하다.
HOLD 소화설비가 있는 선박은 손상여부를 점검하고 CO2 LINE/NOZZLE CLEAN UP한다.

To remove cargo residues from Hatch Cover Frames, Drain Channels, Ventilators, Hold
Accesses etc. In case of the vessel fitted with cargo hold fire-fighting facility, it should be
checked whether there is any damage to it, and CO2 Lines and nozzles should be cleaned.

3) F/WATER RINSING (Rinsing with Fresh Water)

청수는 양하항에서 충분히 확보 후 전 HOLD 를 RINSING 시행

Sufficient quantity of fresh water for cleaning should be supplied at discharge port and all of
the cargo holds should be rinsed with fresh water.

2.2.4 DRY

기상/해수온도/인접 BALLAST TANK의 BALLAST 온도에 의한 CONDENSATION 이나 SWEAT를
고려하여 HOLD 건조 시행하고, HOLD VENTILATOR가 설치된 선박은 적극 활용 한다.

Cargo holds should be dried, considering the weather conditions, temperature of sea water,
any condensation or sweat by adjacent ballast water tanks, etc.

The vessels fitted with cargo hold ventilators should use them for drying the holds.

2.2.5 Cleaning Hold Bilge Well

BILGE WELL 내 찌꺼기 제거, DRY UP 후 화물이 혼입되지 않도록 BURLAPPING 후 고박한다.

Residues in the bilge wells should be removed, and the bilge well should be covered with burlap
after drying so as to prevent the ingress of the next cargo

2.2.6 Chipping/Scraping and Painting

LOOSE SCALE 제거 후 필요 시 도장 시행, 특히 FRAME 뒷면이나 구석진 부위를 철저히
확인 한다.

Loose scales in the holds should be removed, and area painted if necessary. Areas, particularly
behind the frames and corner parts should be checked thoroughly.

2.2.7 Ventilation of Cargo Hold

환기 부적절로 화물창 검사가 FAIL되는 경우가 있으니, PAINTING 한 경우 입항 3-4일전 도장 작업 완료 후 환기 시행하여야 한다. 건조 완료 후에는 VENTILATOR/HOLD ACCESS/H.COVER 등 폐쇄 하여야 한다 (벌레 등의 침입 방지)

In case if painting is carried out, the painting work should be completed at least 3~4 days before the load port and ventilation should be carried out.

Cargo hold inspection can fail due to insufficient ventilation.

After the drying of cargo holds, the ventilators, hold accesses, and hatch covers should be closed(to prevent the ingress of insects, etc.).

2.2.8 Inspection of Ballast System

발라스트 계통의 각종 VALVE, INDICATOR, HYD 계통(HYD PUMP, HYD ISO VALVE, ACTUATOR, EM, CY HYD PUMP) 이상 여부를 확인하고, PIPE 손상 및 누수여부를 확인해야 한다.

BALLAST STRIPPING SYSTEM (PUMP, V/V, PIPE, EJECTOR 등외)

BALLAST PUMP

All valve and indicators

Hydraulic system (Hyd. Pumps, Hyd. ISO valves, Actuator, EM, CY Hyd. Pumps)

Remote sounding gauges

Damage and leakage of every pipe

Ballast stripping system (Pump, V/V, Pipe, Ejector, etc.)

2.2.9 Inspection of Hold Bilge System

- PMS에 의거 BILGE V/V (Non-return vv)개방 정비 및 점검이 되어야 한다.
- MAIN BILGE LINE에 설치된 DRAIN CORK을 통하여 V/V 누설 확인을 하고,
- BILGE PUMP SUCTION 측 GLOVE CHECK V/V선창소재 후 분해/점검해야 한다.
- 선창소재 후 BULK 선 운항 CHECK LIST 의거 BILGE V/V 작동상태 확인 및
- BILGE 배출 후 전 V/V 폐쇄 여부 확인을 하여야 한다.
- 곡물 등과 같이 흡습화물 운송 중 HOLD BILGE가 생성된 적이 있을 경우에는 반드시 차항 화물 선적 전까지 원인을 파악하고 수리 조치하여야 한다.

Bilge valves are opened and maintained according to the PMS.

To check any leakage of valves through the drain cork fitted in the main bilge line.

Glove check valve at bilge pump suction should be overhauled and inspected after hold cleaning

To carry out operational test of bilge valves according to Bulk Cargo Operation Check List after hold cleaning.

To check whether all valves are closed tightly after discharging bilge water.

In case any hold bilge water was generated during the transportation of hygroscopic cargo such as grain and, the cause of such generation should be found out and repaired before loading cargo of next voyage.